Shrey Deshmukh

sdeshmu2@uci.edu | +1 (949) 331-5146 | LinkedIn | GitHub | Irvine, CA

Work Experience

AI Research Intern | ETAD Lab, Calit2 ($300k Grant - Asahi Group Holdings) **June 2026 - Present**

- Building inference for StressDX, a chronic stress detection system running entirely on-device with no cloud offload, classifying chronic stress from Apple Watch and Garmin physiological signals for continuous background monitoring.
- Reduced inference time to milliseconds by implementing Resonator Networks and Neural Vector-Symbolic Architectures in PyTorch and TorchHD, optimizing for performance on resource-constrained devices.

Hewlett Packard Enterprise | Software Engineer - Hybrid Cloud Storage **August 2023 – August 2025**

- Automated provisioning across distributed Kubernetes clusters managing **14PB** of enterprise data by building and containerizing Go microservices with Docker, released through Helm on Jenkins CI/CD pipelines.
- Wrote the **Golang** controllers and **Kubernetes** CRDs behind automatic failover, so workloads keep running when the link between two arrays drops. Shipped across six storage platforms in the GA release of HPE's **open-source** CSI Driver v3.0.0.
- Closed **12+ customer-critical escalations** on **Fortune 500** Kubernetes deployments with zero reopened cases by tracing volume attach and replication failures through Grafana dashboards, customer logs, and API responses.
- Cut datastore recovery time from hours to near-instant by implementing a fast-copy snapshot restore path over array-level primitives, tied to a **$1.1M** disaster recovery engagement.
- Led team adoption of **AI-assisted development** with GitHub Copilot and GoMonkey mocks, raising unit test coverage on owned Go modules from 0% to 80% and cutting test authoring from 3 days to 1.
- Co-founded the HPE Innovation Connect Program, a cross-team program that took 4 engineering teams from hackathon prototype to TechCon submission, with **all 4 winning at HPE TechCon 2025**.

Hewlett Packard Enterprise | Software Engineer Intern - Hybrid Cloud Storage **January 2023 – July 2023**

- Replaced an 8-step manual runbook with one automated workflow and removed 60% of manual recovery checks, measured against support call logs, by building a multithreaded Python and Go site recovery plugin.
- Placed **Top 5 of 200** teams at **HPE's All India Hackathon 2023** by building a guided Go CLI over storage REST APIs, with safety checks across 14+ edge cases.

Projects

Sustainable Product Design Research Agent | Live Good, Inc. USA

Python, FastAPI, LangGraph, GCP, Terraform, Docker, Gemini

- Deployed a containerized service to GCP Cloud Run entirely through Terraform, securing its public endpoint with token auth and least-privilege IAM to prevent unauthorized use and runaway cost.
- Caught quality regressions by wiring 7 log-based metrics and 2 alert policies to the service's evaluation scores, which reached 90-93% citation coverage across 182 sources.

Transaction Processing Engine

C++, Python, RocksDB, Multithreading

- Sustained ~11,000 transactions/sec at 12ms p99 under heavy contention by designing a multithreaded OLTP engine on RocksDB with a hand-written parser and evaluator for a custom transaction language.
- Eliminated deadlocks by design across 120 benchmarked combinations of thread count and contention level by implementing Optimistic Concurrency Control and Conservative 2PL with sorted lock acquisition.

Education

University of California Irvine (UC Irvine) **September 2025 - December 2026**

Master of Computer Science, GPA 3.82/4.0

Coursework: Distributed Systems, Algorithms, Parallel Computing, Advanced Data Structures, Transaction Processing

Manipal Institute of Technology, Manipal **July 2019 - July 2023**

B.Tech Computer and Communication Engineering, Minor in Big Data Analytics

Leadership: Captain, Athletics and Cross Country (led 30+ athletes); Vice President, Astronomy Club (mentored 50+ students)

Technical Skills

Languages: Go, Python, C++, TypeScript, SQL, Bash
Infrastructure & Databases: Kubernetes, Docker, Helm, Terraform, Jenkins (CI/CD), Git, Linux, PostgreSQL, RocksDB
Cloud: GCP (Cloud Run, Cloud Build, IAM), AWS (ELB, ASG, ElastiCache, Bedrock), VMware vSphere (Private Cloud)
AI & ML: LangGraph, Gemini API, Claude API, Copilot, Agentic Workflows, RAG, On-Device Inference, PyTorch, TorchHD, NumPy
Backend & APIs: REST APIs, gRPC, FastAPI, Microservices, Concurrency, Multithreading, Distributed Systems
Testing & Debugging: GoMonkey, Mockito, PowerMock, Unit and Integration testing, Root Cause Analysis, Logging, Code Review
Certifications: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner